

Blockchain

ACADEMIC CERTIFICATES VERIFICATION IN 2 SEC

The project proposes a certificate issuance and verification solution based on blockchain and IPFS technologies that aim to provide decentralised, transparent, secure system for storing, issuing, and verifying academic certificates.

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INTRODUCTION

The process of verifying documents is known to be an expensive and time-consuming activity. In the employment process, an applicant must provide the necessary documentation, and the recruiter is typically tasked with verifying the authenticity of these documents. This involves contacting the relevant institution or enlisting a third-party service provider. The UniCertify project aims to simplify this process by developing a web-based interface to handle both the issuance and verification of documents eliminating the need for contacting the institution or third parties, and thus, reducing both time and cost.

OBJECTIVES

- Streamline the process of document issuance and verification in the employment and academic sectors.
- Develop a simple intuitive blockchain system that is used in securing, storing, and verifying academic certificates, reducing fraud and forgery risks.
- Eliminate the time-consuming and costly traditional methods used in the certificate verification process.
- Provide transparency and accuracy in document verification.

TOOLS USED



IPFS



Metamask



Smart contract



Blockchain



React



JavaScript



Vite



Hardhat



Solidity

thirdweb

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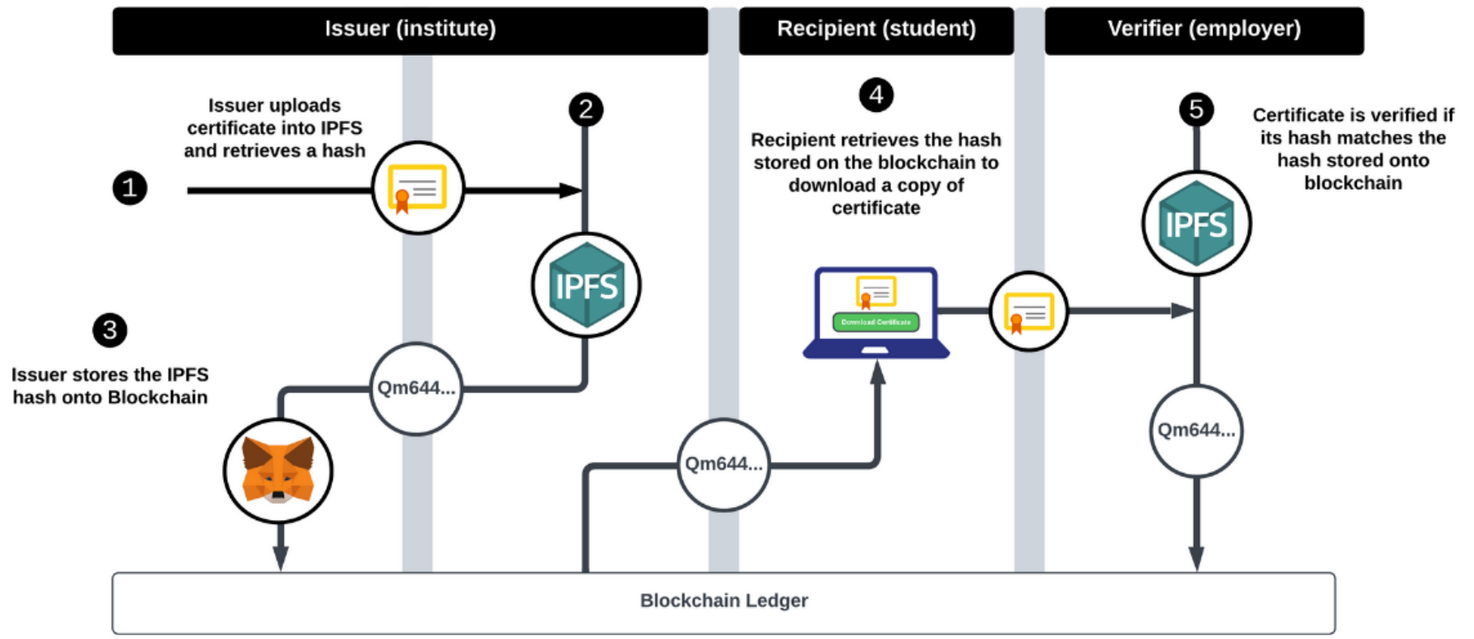
RESULTS/FINDINGS

The project can significantly revolutionise the current certification verification process for numerous reasons including:

- Enhanced security measures, reduced administrative costs, and an increase in transparency.
- Using a decentralised system such as blockchain eliminates the need for third parties and reduces data fraud risks.
- Certificates are immutable and permanently stored in the blockchain with complete accessibility and total independence from the institution

SYSTEM WORKFLOW

Firstly, the issuer uploads the student's certificate to the IPFS network, and accordingly, a 32-bit hash string is generated to the corresponding certificate's content. Once the hash is generated, it's added to the Ethereum blockchain along with the student's information through a MetaMask transaction. Following the certificate issuance, the student receives an invitation from the institution to view his/her certificate portal on the web-based application. The student accepts the invitation to view or download a digital copy of his/her certificate. When applying for jobs, a student can send a digital certificate via email which can be simply verified in seconds through the verification portal free of charge by uploading the provided certificate to the IPFS and generating a new hash. The new hash can then be compared to the hash stored on the blockchain during the issuing process.



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METHODOLOGY

Mixed methods are used in collecting, analysing, and evaluating the project's data. including both qualitative and quantitative methods.

These methods include:

- Surveys
- Comparison studies
- Experiments
- Transaction logs

The development model implemented in this project is the Agile methodology that can adapt to changing situations and requirements through dividing the development process into smaller tasks.

PERFORMANCE

To accurately evaluate the whole system's performance and the amount of gas fees needed to perform transactions, Etherscan will be used to collect data on each transaction in the system.

Value:	0 ETH ($0.00)
Transaction Fee:	0.238135334087691306 ETH $0.00
Gas Price:	207.461525202 Gwei (0.000000207461525202 ETH)

Gas fees consumed for deploying the smart contract.

Value:	0 ETH ($0.00)
Transaction Fee:	0.00090338202789549 ETH ($0.00)
Gas Price:	2.252177126 Gwei (0.000000002252177126 ETH)

Gas fees consumed for uploading a certificate.

Value:	0 ETH ($0.00)
Transaction Fee:	0.00026818305204890 6 ETH ($0.00)
Gas Price:	2.244153302 Gwei (0.000000002244153302 ETH)

Gas fees consumed for assigning an authorised admin.

CONCLUSION

The UniCertify project addresses the significant drawbacks that exist in the current methods of issuing and verifying certificates. The proposed solution revolutionises the certification verification process and tackles certification fraud. The platform is built to be a globally used system in multiple institutions and universities that would be interested in implementing a blockchain and an IPFS system for storing and verifying certificates in their educational systems.